

exactly 80 mm: _____

FIRST SOLVE

New here? Do the walkthrough once at cubepath.com/3x3x3-vercel.app/.
 This code is the memory jog:
NOTATION
 R U L D B = turn that face clockwise, looking at from outside the cube
 r = counter-clockwise, z = half turn. Lowercase r = that face plus the layer behind it, turning together.
 y = spin the whole cube left, white staying down. A whole-cube turn follows nothing.
WORDS
 algorithm = a fixed sequence of turns - trigger = a short algorithm your hands run as one unit
1 WHITE CROSS



White center DOWN. Bring the four white edges to it. Each edge's side color must match the center it touches. No algorithm - work it out. Can't see it? Build the four white edges on the YELLOW face first, then solve its matching center, then turn each one down 180 degrees.

2 WHITE CORNERS

Find a white corner on top, turn U until it sits above the yellow slot it belongs in, then match a picture below and repeat until it drops in.



white sticker RIGHT
RU'R



white sticker FRONT
LU'L



white sticker UP run the first one once to knock it out, then look again
RU'R

3 MIDDLE EDGES

Top edge with NO yellow: turn U until its front color matches the center it belongs to. The arrow shows which slot it goes to.



goes **RIGHT**
U RU'U' y LU'LU



goes **LEFT**
U' LU'LU y' RU'RU'

Edge stuck in the wrong slot? Run either one to kick it out, then place it.

■ RU'R sexy move
 ■ LU'L Suno
 ■ RU'FRF' sledgehammer
 ■ gap = change grip

FIRST SOLVE

New here? Do the walkthrough once at cubepath3.sixvercel.app/ learn. This card is the memory jogy.

NOTATION

R U F L D B = turn that face clockwise, looking at it from outside the cube
 r counter-clockwise, 2 = half turn
 Lowercase r f = that face plus the behind it, turning together
 y = spin the whole cube left, white staying down. z = whole-cube turns below nothing.

WORDS

algorithm = a fixed sequence of turns = trigger = a short algorithm your hands run as one unit

1 WHITE CROSS



White center DOWN. Bring the four white edges to it. Each edge's s color must match the center it touches. No algorithm – work it out. Can't see it? Build the four white edges on the YELLOW face first, each above its matching center, then turn each one down 180 degrees.

2 WHITE CORNERS

Find a white corner on top, turn U until it sits above the empty slot it belongs in, then match a picture below and repeat until it drops in.



white sticker RIGHT
RUR'U'



white sticker FRONT
L'ULU



white sticker U run the first one once to knock it out, then look again
RUR'U'

3 MIDDLE EDGES

Top edge with NO yellow: turn U until its front color matches the center under it. The arrow shows which slot it goes to

goes RIGHT
U' RUR'U' y L'ULU



goes LEFT
U' L'ULU y' RUR'U'

Edge stuck in the wrong slot? Run either one to kick it out, then place it.

RUR'U' sexy move **RUR'U** Sune **R'FRF** sledgehammer gap = change grip

TWO-LOOK

OLL CROSS - yellow edges



Line bar left-to-right
fRUR'U'f



Hook hook pointing front-right
fRUR'U'f

OLL CORNERS - yellow face



Sune $\frac{1}{2}$ yellow, rest clockwise
RUR'U RU2R'



Anti-Sune $\frac{1}{2}$ yellow, rest counter-clockwise
RU2R' U'RUR'



P1 $\frac{1}{2}$ yellow; headlights left
fRUR'U'f fRUR'U'f

OLL CORNERS - continued



Headlights $\frac{3}{4}$ 2 yellow; headlights at the back
R2DR'U2 RD'R'U2R'



2x Headlights $\frac{3}{4}$ 0 yellow; headlights both sides
RUR'U RU'R' URU2R'



Chameleon $\frac{3}{4}$ 2 yellow; next to each other
rUR'U r' FRF



Bowie $\frac{3}{4}$ 2 yellow diagonal
fR'UR'U r' FR

FAILBACK

The badge on each row is your failback: that many Sunes, with a U turn around, also finishes the case. Machine-checked, there is the worst.
Top not all yellow and nothing matches? Run Sune and read again.

No yellow edge at all? Run the first one, then read again.

■ RUR'U sexy move
 ■ RUR'U Sune
 ■ R'FRF sledgehammer
 gap = change grip

TWO-LOOK

OLL CROSS - yellow edges



Line bar left-to-right
fRUR'U'f'



Hook hook pointing front-right
fRUR'U'f'

OLL CORNERS - yellow face



Sune $\frac{1}{2}$ yellow, rest clockwise
RUR'U RU2R'



Anti-Sune $\frac{1}{2}$ yellow, rest counter-clockwise
RU2R' U'RUR'



Pi $\frac{1}{2}$ yellow; headlights left
fRUR'U'f' fRUR'U'f'

OLL CORNERS - continued



Headlights $\frac{3}{4}$ yellow; headlights at the back
R2DR'U2 RD'R'U2R'



2x Headlights $\frac{3}{4}$ yellow; headlights both sides
RUR'U RU'R' URU2R'



Chameleon $\frac{3}{4}$ yellow; next to each other
fRUR'U'f' FRF'



Bowtie $\frac{3}{4}$ yellow diagonal
f'RUR'U'r' fR

FALSEBACK

The badge on each row is your falseback: that many Sunes, with a U turn between, also finishes the case. Machine-checked, there is the worst.
Top not all yellow and nothing matches? Run Sune and read again.

Only no yellow edge at all? Run the first one, then read again.

■ RUR'U scwy move ■ RUR'U Sune ■ R'FRF' sldchangearm gap = change grip

ONE-LOOK PLL

ADJACENT CORNER SWAP exactly one face shows headlight



***T.L. headlight; pairs R-left F-left**
RUR'UR' R'F R2UR'UR' RUR'F'



Ja L solid; pairs B-right F-left R-front
x R2FRF' RU2 rUr U2



Jb L solid; pairs B-left F-right B-back
RUR'F' RUR'F' R'F R2UR'F'



Aa L solid; pairs F-right R-front
x L2D2 L'U'L D2 L'U'L



Ab L headlight; pairs B-right R-back
x L2D2 LUL' D2 LU'L



F.L solid
RUR'F' RUR'UR' R'F R2UR'UR' RUR'UR'

One face shows headlight; two corners of that face match. Hold as shown, run, then turn the top face to finish.

RUR'F' sexy, moves, B.RUR'F' sync, R'FRF' sldochanger, ego - channie erin

ADJACENT CORNER SWAP continued



Ra L. headlight; pairs R-left
RUR'UR' RUR DR'UR'D' R'U2R'



Rb L. headlight; pairs R-left
R2F RUR'UR' F'R U2R'U2R



Ga L. headlight; pairs F-right
R2UR'UR'UR' RU'R2 U'D R'UR'D'



Gb L. headlight; pairs R-back
RUR' UD' R2UR'UR' RU'R2D



Ge L. headlight; pairs R-right
R2UR'RU RU'R2 UD' RU'R'D



Gd L. headlight; pairs R-front
RUR' UD' R2UR'UR' UR'UR2D

Learn in the printed order, three a week, about six weeks. Ja and Jb on the same day – one is the mirror of the other.

ONE-LOOK PLI

ADJACENT CORNER SWAP

exactly one face shows headlight



T L headlights; pairs R-left F-left
R'U'R' R'F R2U'R' R'UR'F'

J L solid; pairs B-right F-left R-front
x R2FRF' RU2 r'Ur U2

Jb L solid; pairs B-left F-right B-back
R'FR' R'UR' R'F R2U'R'

A L solid; pairs F-right B-front
x L2D2 L'U'L D2 L'U'L'

A L headlights; pairs B-right B-back
x' L2D2 LUL' D2 LU'L

F L solid
R'U'F R'UR' R'F R2U'R'U'R'

One face shows headlights. Two corners of that face match. Hold as drawn, now, then turn the top to finish.

R'UR' R'U'F'RU'F' R'UR' R'FRF' slicechange **can = change grip**

ADJACENT CORNER SWAP

continued



R L headlights; pairs F-left
RU'R' R'UR DR'UR'D R'U2R'

Rb L headlights; pairs B-left
R2F R'UR'U'R' F'R U2R'U2R

G L headlights; pairs F-right
R2UR'UR'U'R' RU'R2 U'D R'UR'D'

Gb L headlights; pairs B-back
R'UR' UD' R2'UR'UR'U'R' RU'R2D

G L headlights; pairs B-right
R2'UR'UR'U'R' R'UR'2 UD' R'UR'D'

Gd L headlights; pairs R-front
RUR' U'D R2'UR'UR'U'R' UR'UR'D

Learn in the printed order, three a week, about six weeks. Ja and Jb on the same day – one is the mirror of the other.

ANALYSIS - BIG CUBES and THE MAP

BIG CUBES 4x4 and 5x5
 Reduce: build the centers → pair the 12 edges → solve it as a 3x3. Rw = 2 layers wide. 3Rw = 3 layers. 2R = 2nd Layer ONLY → never widen it.
 last 2 edges
 4x4 PLL parity
 4x4OLL parity → 2 edge pair flipped - half the time
 Rw U2 x Rw U2 Rw U2 Rw U2
 Rw U2 x Rw U2 Rw U2 3Rw U2
 U2 Lw U2 Rw U2 Rw U2 Rw U2 Rw U2
 U2 Lw U2 Rw U2 3Rw U2
 Fix parity during the last layer, before OLL and PLL - both parity algorithms move corners.
NOTATION
 Each letter turns that face clockwise, seen from your outside the cube → so from your seat, D and B look clockwise. → counter-clockwise = 2 half turn.
 M = middle slice, turning the way L turns. x y z = the whole cube, turning the way U and F turn. Lowercase f = that face plus the layer behind it, turning together.
WORDS
 algorithm → a fixed sequence of turns - trigger = a short algorithm your hands run as one unit - sexy move = the R U R' U' trigger -
 sledgehammer = the R' F R' trigger - parity = a case a 3x3 cannot have - edge pair → two edges glued into one, on a 4x4 or 5x5
 R U R' sexy move R U R U S R' F R' sledgehammer gap = change grids

slice out, run it, slice back - both cubes
 2 edge pairs swapped - half the time
 one corner flipped, 3 has the PLL parity
 use it until you know Sure
 turn a yellow corner four turns at a time, turning ONLY the top face between corners.
 yellow faces RIGHT R'D'R D
 yellow faces FRONT R'D'R D x2

ANNEX – BIG CUBES AND THE MAP

BIG CUBES 4x4 and 5x5

Reducing a build the 6 centers → pair the 12 edges → solve it as a 3x3. Rw = 2 layers wide. 3Rw = 3 layers. 2R = 2nd LAYER ONLY – never widen it.

widest edges

Rw'	U'	R'	U'	R'	F	R'	F'	Rw	slice out, run it, slice back – both cubes
2R2 U	2R2 U	2R2 U	2R2 U	2R2 U	2R2 U	2R2 U	2R2 U	2R2 U	Z edge pairs swapped - half the time

4x4 PLL parity

4x4 OLL parity = 1 edge pair flipped - half the time

Rw	U2 x	Rw	U2 R'w	U2 R'w	U2 R'w	U2 R'w	U2 R'w	U2 R'w	U2 R'w	U2 R'w	U2 R'w	U2 R'w	U2 R'w
Rw	U2 x	Rw	U2 R'w	U2 R'w	U2 R'w	U2 R'w	U2 R'w	U2 R'w	U2 R'w	U2 R'w	U2 R'w	U2 R'w	U2 R'w

Twist a yellow corner four times at a time, turning ONLY the top face between corners.

yellow faces RIGHT **R'D'R'D x2**

yellow faces FRONT **R'D'R'D x2**

NOTATION

Each letter turns that face clockwise, seen from outside the cube – so from your seat L, D and B look counter-clockwise. = counter-clockwise, z = half turn.

M = middle slice, turning the W layer.
Turn x y z corner = the whole cube, turning the way R, U and F turn. Lowercase r f = that face plus the layer behind it, turning together.

WORDS

algorithm = a fixed sequence of turns + trigger = a short algorithm you hands run as one unit - sexy move = the R U R' U' trigger - slicechange = the R' F R' F' trigger, parity = a case a 3x3 cannot have; edge pair = two edges glued into one, on a 4x4 or 5x5

RURU sexy move **RURUSe** slice **R'F'RF** slicechange **g**= change grip

Print at 100% / Actual size — do NOT fit to page. Two-sided, flip on the SHORT edge. The number of pages is 10. The width of the page is exactly 80 mm: _____

[illegible]